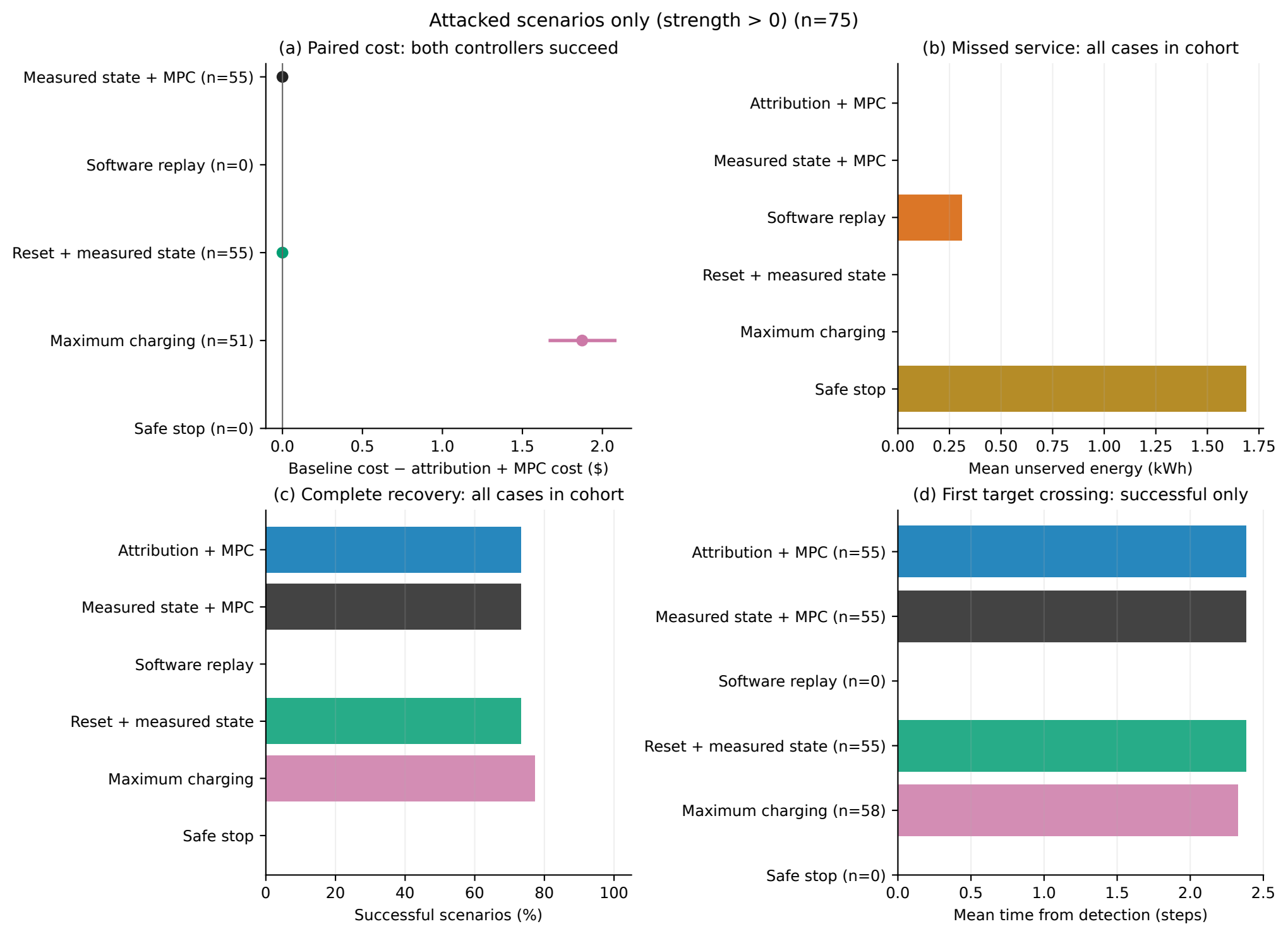
Attacked scenarios only (strength > 0) (n=75)
(a) Paired cost: both controllers succeed
Measured state + MPC (n=55)
Software replay (n=0)
Reset + measured state (n=55)
Maximum charging (n=51)
Safe stop (n=0)
0.0
0.5
1.0
1.5
2.0
Baseline cost − attribution + MPC cost ($)
(b) Missed service: all cases in cohort
Attribution + MPC
Measured state + MPC
Software replay
Reset + measured state
Maximum charging
Safe stop
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Mean unserved energy (kWh)
(c) Complete recovery: all cases in cohort
Attribution + MPC
Measured state + MPC
Software replay
Reset + measured state
Maximum charging
Safe stop
0
20
40
60
80
100
Successful scenarios (%)
(d) First target crossing: successful only
Attribution + MPC (n=55)
Measured state + MPC (n=55)
Software replay (n=0)
Reset + measured state (n=55)
Maximum charging (n=58)
Safe stop (n=0)
0.0
0.5
1.0
1.5
2.0
2.5
Mean time from detection (steps)